Supernature: the plan

Build Queue item 96 · Claudia · October 13, 2026 · **Status: planned, not built.** All four of her answers are in (2026-10-13, section 12). Building starts when her assets arrive.

Supernature is a new app on the student computer. It replaces the old idea of a "Potion Lab". Students discover real mythical creatures from around the world, make their own creatures, and build a small living world where those creatures have to eat, rest and get along. Potions and wand spells are the magic that ties it together. Everything is 2D pixel art. Nothing is shared between students. **There is no blood anywhere: not in Supernature and not in any other part of the platform.**

Her words that set the direction (2026-10-13): "creating student own supernatural creatures and their interaction with the ecosystem the student creates seems to be the priority. these are ideas from students, claudia should explore connections, patterns, and actual game play logic." Then: "claudia, make sure this is a cohesive, dev plan before going. ill get you assets."

1. The one big idea

Supernature works like a field naturalist's journal that happens to be magic. Every part feeds the next part, so a student always has a reason to go somewhere else in the app:

Room	What the student does	What it gives the other rooms
Globe	Spins the world, taps a continent, ocean or sea, zooms in to countries, finds glowing creature pins	Unlocks Bestiary pages, earns ingredients ("Earned coral by exploring the Mermaid")
Bestiary (the creature book)	Reads each creature's page; unknown creatures are "???" silhouettes; sorts trait cards to finish a page	Real creatures to remix; trait words to build with
Creature Maker	Explores, remixes, or builds a creature from scratch out of pixel parts and trait cards	Creatures to put in a Habitat
Habitat (the priority)	Picks a biome, adds plants, real animals and creatures, runs the seasons, keeps the food web in balance	Adaptation traits, creature levels, Naturalist rank
Cauldron	Reads a recipe and follows it in order, mixes, then traces the spell word in cursive with a wand	Potions for pets, for their own character in Town Square, and for Habitat problems
Supernature Shop	Buys ingredients with their real dashboard money	Ingredients for the Cauldron

The loop a student lives in: **find it, learn it, make one, keep it alive, brew for it.**

2. Learning goals and where each one lives

- **Mythology and culture** (Globe, Bestiary): real creatures, the country and people whose stories they come from, told respectfully (section 9).

- **Ecosystems are fragile** (Habitat): producers, herbivores, carnivores, omnivores and decomposers; food chains and the food web drawn as arrows; what happens when one part is removed.
- **Geography** (Globe): continents, the five oceans and big seas, then countries.
- **Vocabulary through sorting traits** (Bestiary, Creature Maker, Habitat): her list, grouped so students sort them:
  - Body: sharp, sword-like hands, one eye, legs, flippers, wings.
  - Moving: walking, flying, swimming (aquatic), burrowing.
  - Survival: water breathing, regenerative, immunity to pain, endurance (the cockroach is the champion).
  - Mind: an intelligence scale (from "follows its senses" up to "makes fire and tools"), mischievous, curious, gentle.
  - Social: pack hunter or loner, cooperation.
  - Adaptations: traits learned from where the creature lives (section 5).
- **Reading is a job** (Cauldron, Bestiary): every recipe and creature page is read; read-aloud on every screen.
- **Order of operations** (Cauldron): recipe steps happen in order, and anything in the small mixing bowl happens first, just like parentheses in math (section 6).
- **Cursive** (Wand): the class has learned every letter and is writing words, so students write whole words in Cursive Without Tears letter shapes, mostly creature names (section 7).
- **Energy language** (everywhere): Energy check-ins on their creatures, using the 5-step energy scale with the original faces from her MyEnergy page (section 8). This is language the class is building, so it runs through every room.

Questions from question sets are **not** part of Supernature for now (her answer: "no questions yet").

3. Globe

- A pixel-art globe students spin with a swipe and zoom with a pinch. It turns slowly on its own when nobody is touching it.
- **Zoomed out:** the seven continents, the five oceans (Pacific, Atlantic, Indian, Southern, Arctic) and the big seas (Mediterranean, Caribbean, South China Sea, North Sea and a few more). The one under a finger lifts and glows (touch, not hover, since students are on iPads). Its name is said aloud.
- **Zoomed in:** a flat pixel map of that continent or ocean with country borders. A country with a discovered or discoverable creature has a pin. Pins for creatures not found yet show "?".
- Tapping a pin opens the Bestiary page for that creature. The first visit earns that creature's ingredient.
- **Antarctica** has no traditional creature stories, because no people lived there to tell them. That fact is taught on the page, and it is the open frontier: students may place their own creatures there.
- Built from public domain world map data (Natural Earth), drawn down to a pixel grid, so borders are real.

4. Bestiary (the creature book)

- A pixel spell book with page turns, like her reference picture. Pages are grouped by continent and ocean, then a **My Creatures** section at the back.
- An undiscovered creature is a dark silhouette with "???" and only its region shown.
- A discovered page shows: the pixel picture, its name and how to say it (read aloud), where it comes from (country and people), a short story told respectfully, its traits, its diet, its habitat, and its usual energy ("Kappas are often at high energy: they love mischief.").
- **Write its name in cursive:** the student writes the creature's name with the wand on a cursive line (section 7). The name is then shown on the page in their own handwriting.
- **Finish the page job (vocabulary):** trait cards are mixed up beside the page; the student drags each to "has it" or "does not have it", using the story they just read. A finished page gets a gold seal. Kind help on a wrong sort, never a buzzer.
- Story pages read themselves aloud by default with a 🔊/🔇 toggle (her rule for story screens).

5. Creature Maker and Habitat (the priority)

Creature Maker

Four ways in, her list:

1. **Explore a real creature:** read it and see it move. Read only.
2. **Remix a real creature:** start from a real one and swap parts and traits. The card says "Inspired by the Griffin (Greece)". Only creatures marked "remix ok" can be remixed (section 9).
3. **Build from scratch:** pick a body plan (walker, flier, swimmer, slitherer, floater), then pixel parts: head, eyes (1, 2 or 3), arms (hands, claws, sword-hands, flippers, tentacles), covering (fur, scales, feathers, shell, slime), tail, wings, colors.
4. **Spark:** a random start for students who freeze at a blank page.

Then the student:

- Sees **body traits filled in automatically** from the parts (three eyes gives "sharp sight"; flippers give "aquatic"). These are the traits "assigned to the creature".
- **Chooses personality** (mischievous, gentle, curious, bossy, shy), **social style** (pack or loner), **intelligence** (1 to 5) and **diet** (plants, meat, both, leftovers, or a magic food like moonlight or dreams).
- **Names it in cursive** with the wand (section 7), then writes one sentence about it. The sentence check is gentle: a capital letter at the start, punctuation at the end (the same habit as the gym form).
- **Picks its usual energy** on the 5-step scale ("My creature is usually at low energy. It likes to nap in the sun.").
- The creature goes into My Creatures in the Bestiary.

Habitat: the actual game play logic

This is where students' creatures interact with a world they made.

1. **Pick a biome:** forest, ocean, desert, mountain, swamp, tundra, grassland, or sky islands. Each biome has sun, water and cold levels.
2. **Place life:** producers (grass, trees, kelp, plankton, glowing mushrooms), real animals (rabbits, deer, fish, beetles, cockroaches), and up to a few of their creatures. Each is a pixel sprite that walks, swims or flies around.
3. **Run a season:** the student taps "Next season". Each season:
    - Producers grow, depending on the biome's sun and water.
    - Herbivores eat producers; carnivores eat herbivores; omnivores eat either; decomposers turn leftovers back into soil, which helps producers.
    - Each kind's group gets bigger, stays the same or gets smaller, shown as pixel counters (1 to 10 icons).
    - **Eating is never shown.** A predator "catches" and a little poof of smoke shows the prey running back to its burrow; the number goes down. A group that runs out of food "packs up and moves away" (a tiny suitcase walking off screen). No blood, no hurt animals, ever.
4. **The food web** is a tap away: arrows from each food to whatever eats it (arrows show where the energy goes). Students can turn on "what if": hide one kind and watch a ripple show who is affected. This is the fragility lesson.
5. **Balance meter:** Balanced, Wobbly or Falling apart, with one plain sentence saying why ("Too many hunters and not enough rabbits.").
6. **Traits matter:** a creature without water breathing struggles in the ocean; pack hunters catch bigger food but need more of it; regenerative creatures bounce back after a hard season; endurance (the cockroach trait) lets a kind survive a bad season; a fire-making creature can warm the tundra. Students learn which traits fit which places by trying.
7. **Adaptations are learned:** a creature that lives through three seasons in a biome may learn one of that biome's adaptations (thick fur in the tundra, night vision in a cave, camouflage in the forest). The student picks the one that fits; it is added to the creature's page as "learned".
8. **Energy check-ins:** every creature living in the Habitat, real ones they found and ones they made, shows energy clues as it moves: zooming in circles, bouncing, walking calmly, dragging slowly, yawning. Each season the student does an Energy check-in on a creature that needs one (section 8). Energy is part of the balance: crowded, overfed or overplayed creatures run high; hungry or lonely ones run low.
9. **Levels:** creatures gain a level after each balanced season with their energy at just right. The student earns a **Naturalist rank** (Apprentice, Field Scout, Creature Keeper, Habitat Builder, Grand Naturalist) that unlocks new biomes, parts and recipes.

Saved per student and private: their creatures, habitats, discovered pages, ingredients and potions.

6. Cauldron (potions)

- **The Cookbook:** about 10 base recipes to start (her number). Ideas: Flying, Strength, Speed, Glow, Tiny, Giant, Calm, Zip, Bounce, Grow (for plants in a Habitat).
- **Discovered recipes:** brewing a base recipe unlocks a harder version with bigger needs: Flying unlocks Super Speed Flying; Glow unlocks Rainbow Glow; Strength unlocks Mountain Strength.
- **Reading is the job:** the student reads the recipe card (read aloud on tap) and follows it.
- **Order of operations:** each recipe is a list of steps: add, stir (a number of times), heat, wait, wand. Anything in the **small mixing bowl** happens first, like parentheses. Example: "Mix (grass + dew) in the bowl. Pour it in. Add the feather. Stir 3 times. Trace *fly*." Doing a step out of order is never a fail: the cauldron fizzles with a silly puff of smoke and a burp, the step list shows which step comes first, and they try again.
- **Big magic effects:** smoke poofs, sparkles, bubbling, color swirls, a lid that rattles. Calm mode and reduced motion turn these down.
- **Where potions go:**
  - **Pets:** a Strength potion makes a pet glow and lift a tiny dumbbell, and it also counts toward gym training to a degree (her answer): +1 training for the trick the pet is working on (a full gym session gives +2), once a day per pet. Speed works the same way. Potions help training along; they never replace the gym or pet care.
  - **The student's own character:** a Flying potion lets their Town Square character float and fly for a minute; Tiny, Giant, Glow and Bounce work the same way.
  - **Habitats:** Grow helps a struggling plant group; Calm brings a high-energy creature down toward just right; Zip brings a low-energy creature up.

7. Wands and cursive

Her answer (2026-10-13): "All letter have been taught. We are onto writing words. Writing the names of the creatures is a good place to have them write."

- **Creature names are the main writing job.** Every time a student discovers a real creature, and every time they name their own, they write the name in cursive with the wand (a finger or an Apple Pencil). Sparkles follow the stroke. The name then appears in their handwriting on the Bestiary page and above the creature in the Habitat.
- **Spells are words too:** the last step of every recipe is writing the spell word (*fly*, *glow*, *bounce*, *super speed*).
- Letters follow **Cursive Without Tears** shapes and stroke order, with connected letters.
- **Help fades as they grow** (the same fade ladder as Grammar Gus): first the whole word shows faintly to write over, then only start dots and a baseline, then just a blank line. A student can always tap Show me to see the word written once, slowly.
- The check is gentle and counts close enough: each letter is checked for its shape and direction, and a letter that needs another try glows so the student knows exactly which one. Never a buzzer.

- Practice on its own: the Cauldron's spell book lets students write any discovered creature name again.

8. Energy check-ins for creatures

Her answer (2026-10-13): "absolutely use a 5-step energy scale with the original faces. Incorporate this concept throughout this in supernature as it is language we are trying to develop. It should be Energy check-ins for creatures they've found or created living in their ecosystem."

The scale uses the original faces from her MyEnergy page (Autism Level UP!), saved in the project at `public/games/supernature/energy/`:

Step	Face file	Words used everywhere
5	`very-high.png` (frazzled)	very high energy
4	`high.png` (excited)	high energy
3	`just-right.png` (calm smile)	just right energy
2	`low.png` (droopy)	low energy
1	`very-low.png` (yawning)	very low energy
rest	`asleep.png`	asleep (shown when a creature naps or at night; not a step)

The check-in happens on creatures, never on the student: creatures they found or made that live in their Habitat.

1. A creature that needs a check-in gets a small glowing face above it. The student taps it.
2. **Notice the clues:** the creature acts out its energy (zooming, bouncing, calm walking, slow dragging, yawning), and a clue line is read aloud ("Griff is zooming in circles and can't sit still.").
3. **Pick the face:** the 5 faces stand in a column, highest at the top, like her page. The student taps one.
4. **Say it:** a sentence frame builds the language: "Griff has **high** energy." It is read aloud, and the student can tap 🔊 to hear it again.
5. **Pick a tool** to help it get to just right: for high energy, quiet space, a slow walk, a Calm potion; for low energy, food, a friend, play, a Zip potion. The creature tries it, and its face moves toward just right.
6. A wrong pick is never wrong out loud: the clue is shown again with the matching face gently lit.

Where the energy language shows up everywhere else:

- **Bestiary:** each page says the creature's usual energy.
- **Creature Maker:** students choose their creature's usual energy.
- **Habitat:** energy is part of the balance, and creatures level up when their energy is just right.
- **Cauldron:** Calm and Zip potions move energy, and each recipe card says which way it moves it.
- **Host lines** use the same words ("Your Kappa has very high energy. What could help?").
- Wording stays exactly the same every time (very high, high, just right, low, very low), so the words become familiar.

Zones of Regulation stays where it already lives in the dashboard; Supernature uses the energy scale so the two never compete.

9. Culture and respect: the rules and locks

Her rule: "prioritize culturally responsive teaching practices and add parameters and locks around culturally appropriative game play."

- **She approves every region's list** in a teacher screen before any creature appears to students. Each creature has three settings: **Hidden**, **Explore only**, or **Remix ok**.
- **Explore only** is the default for figures that are sacred in a living religion or belong to Indigenous peoples. Students read and learn about them; they cannot remix them, brew from them or put them in a Habitat.
- **Remix ok** is for creatures from old stories that are now shared widely (the Griffin, the Kraken, the Jackalope) and for gods and goddesses of ancient religions no longer practiced as a living faith (Greek, Norse, Roman, ancient Egyptian). She asked to keep gods and goddesses; this is how they stay in respectfully.
- **Left out entirely:** creatures a community has asked outsiders not to tell (the Skinwalker and the Wendigo), and creatures whose story is about blood (the Chupacabra and vampires).
- Every page names the country and the people the story comes from, uses the creature's own name first, and says "In stories told by...", not "This creature is real" or "This creature is fake".
- Ingredients earned from a creature are respectful (coral from the Mermaid, a lost feather from the Griffin); nothing is taken from sacred figures.
- **No sharing:** no gallery, no sending, no showing other students' creatures. She can see every student's book from the teacher side.

Starter creature list: one per clickable region

Her answer (2026-10-13): "start with one in each clickable region in the map. Start with continents and ocean." So the Globe opens with the seven continents and the five oceans, and each region starts with one creature. Seas and countries come later, once the first set works. Proposed picks (she can swap any):

Region	First creature	From	Setting
Europe	Griffin	Greece	remix ok
Asia	Kappa, mischievous and aquatic	Japan	remix ok
Africa	Sphinx	Egypt	remix ok
North America	Jackalope	Wyoming, USA	remix ok
South America	Curupira, forest guardian with backward feet	Brazil	explore only
Oceania	Bunyip	Australia (Aboriginal	explore only

Region	First creature	From	Setting
		peoples)	
Antarctica	none: no people lived there to tell stories, so it is the students' frontier for their own creatures		
Atlantic Ocean	Selkie	Scotland and Ireland	remix ok
Pacific Ocean	Umibozu, the sea giant	Japan	remix ok
Indian Ocean	Mermaid (sailors' stories)	worldwide	remix ok (her example: "Earned coral by exploring the Mermaid")
Arctic Ocean	Kraken	Norway	remix ok
Southern Ocean	none: like Antarctica, the students' frontier		

The rest of the earlier list waits for the next round, still under her approval: Pegasus, Cyclops, Hydra, Unicorn, Loch Ness Monster, Fenghuang, Qilin, Baku, Haetae, Anansi, Bennu, Mokele-mbembe, Thunderbird, Sasquatch, Iara, Alicanto, Taniwha, Rainbow Serpent, Drop Bear, Charybdis, and the gods and goddesses (Zeus, Athena, Poseidon, Thor, Odin, Freya, Ra, Bastet, Anubis), with the same settings as before.

10. Ingredients and the Supernature Shop

- **Free basics:** glass bottle, grass, water, dew, pebble, salt, a feather, moonlight, a leaf, sand. Never gross.
- **Shop ingredients ($1 to $10):** eye of newt (fun fact on the label: in old recipes this was really a mustard seed), hair of the dog, toad burp, pixie dust, troll toenail, dragon sneeze, frog giggle, bat whisper. Gross humor is allowed here but kept to a few (her rule: limited, never in the basics).
- **Super rare ($15 to $50):** phoenix ember, kraken ink, unicorn sparkle, star sap, thunder in a jar.
- Paid with their real dashboard money, the same money as the Marketplace, with a receipt line in their bank history. Prices show with commas.
- **Earned by exploring:** each real creature's page gives one ingredient the first time ("Earned coral by exploring the Mermaid").

11. Other changes that come with it

- **Science Lab:** the fantasy or magic mode is removed; it becomes chemistry only (her rule: "drop fantasy mode").
- **Alchemy:** stays exactly the same.

- **Computer app:** a new "Supernature" icon on the student computer. The host character is to be decided; she is sending the art.
- **Every screen** has read-aloud; story pages read themselves aloud by default with a toggle; voices go through the human voice service.
- **iPad first:** swipe and pinch on the globe, big drag targets, no hover-only actions, both orientations, nothing overlapping.
- **Bug and idea Report** bubble in the corner, like every game.

12. Her answers (all in, 2026-10-13) and what is still owed

1. **Starter list:** one creature in each clickable region, continents and oceans first (section 9).
2. **Cursive:** every letter has been taught; students are writing words; creature names are the writing job (section 7).
3. **Energy:** the 5-step energy scale with the original faces, used throughout, as Energy check-ins for creatures in their Habitat (section 8).
4. **Potions on pets:** count toward gym training to a degree: +1 training, once a day per pet (section 6).

Still owed (she is sending them):

1. The host character (any pose; a few expressions if possible).
2. Pixel art for real creatures, if she has a pack. If not, Claudia builds them from the same pixel parts the Creature Maker uses.
3. Optional: a pixel book and page set, cauldron, wand, potion bottles, biome tiles, smoke and sparkle effects. Anything missing gets drawn in the same pixel style.

13. Build order (each step ships on its own)

1. **Foundation:** Science Lab loses fantasy mode; Supernature app on the computer; saved data; the teacher approval screen for creatures (the 10 starter creatures start approved, everything else Hidden until she approves); the energy faces wired in.
2. **Creature Maker and Habitat** (her priority), with the 10 starter creatures to explore and remix, Energy check-ins, and cursive naming.
3. **Bestiary:** the pixel book, "???" pages, the trait sorting job, My Creatures.
4. **Globe:** spin, zoom, continents, oceans and seas, countries, pins, earned ingredients.
5. **Cauldron and Shop:** the Cookbook, order of operations, discovered recipes, effects, the ingredient economy.
6. **Wand cursive** spells.
7. **Potions out in the world:** pets, the student's Town Square character, Habitats.
8. **Level up** threaded through all of it (started in step 2, finished here).